

Cliff Mass Weather Blog

[This blog provides updated forecasts and comments on current weather or other topics](#)

Friday, July 13, 2012

Thunderstorm Fest (Updated 12:30 PM)

We have had quite a bit of thunderstorm activity over the Northwest the past week and today is no different...in fact, there is continuous thundering outside right now and Seattle is about to be hit!

Last night an usual and spectacular display hit northern Kitsap to the northern Olympics. Here is an amazing picture supplied by Greg Johnson in Hansville. You can see virga (rain) falling out of the cloud near the lightning.



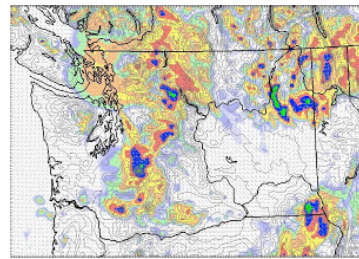
The current radar image, shown below, shows thunderstorms over the north Sound. More convection is over the northern portions of eastern Washington and the coast.



Why the thunderstorms? We start with fairly unstable air above the surface. In fact, you could see it destabilizing during the evening last night with the formation of altocumulus castellanus clouds (castellanus indicating "castles" in the sky!):



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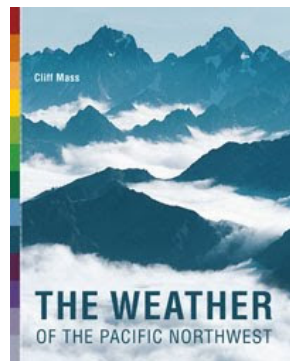


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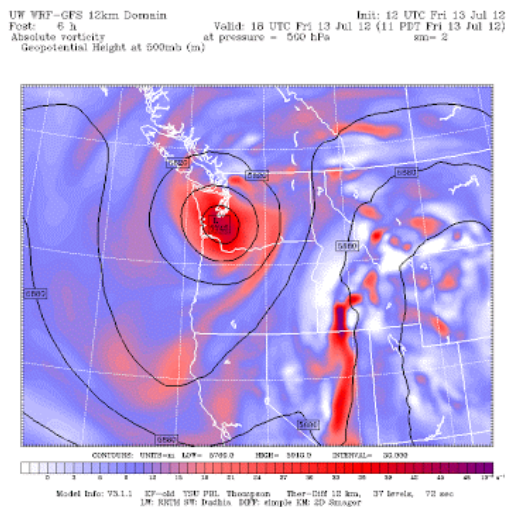
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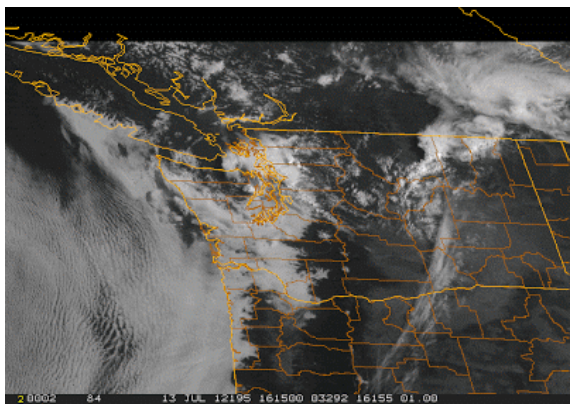


[Click on Picture for More Info](#)

A very tight closed low has moved into the region, with the center near Olympia (see graphic at 500 hPa...around 18K feet) that is rotating moisture northward in eastern Washington, eastward across the northern Cascades and NW WA, and then southward over the Olympics. It is also providing some lift. The key is NOT to have onshore flow through depth...that is death for summer thunderstorms west of the Cascades.

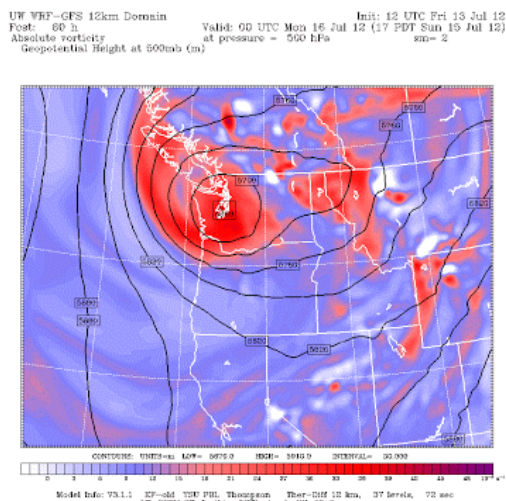


The low center also drove in more marine air at low levels...so quite a bit of lowland clouds this morning (see visible satellite image). It will burn off in most locations later.



The low will move out tomorrow, but the threat of thunderstorms remain...particularly over the mountains and eastern WA.

On Sunday a strong trough moves in that should cool us down 5-10F and bring in more low clouds. The Sunday pattern remains me of the one that produces snow in December...



OK...the perfect weather couldn't last forever...but it will be back by mid-week.

UPDATE:

Here is something fascinating. UW Atmospheric Sciences Grad Student Luke Madaus collected recent Twitter "tweets" that mentioned thunder for a few hours this morning and plotted the density of such messages on a map. Here it is. Lots of reports of Puget Sound and NW Washington!

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- [Windstorm Talk at City Hall](#)

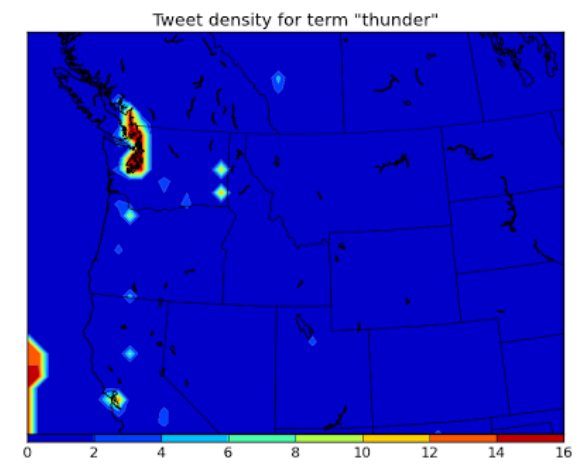
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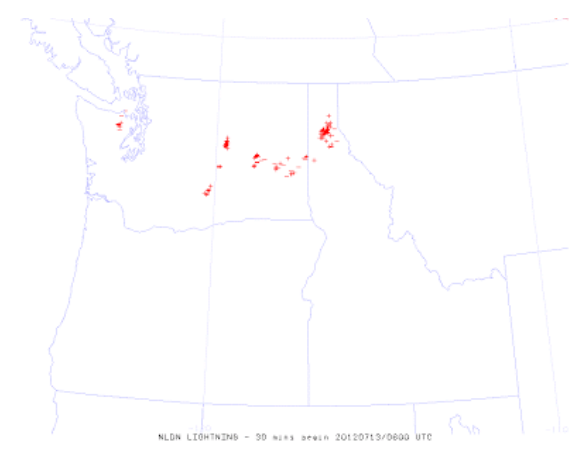
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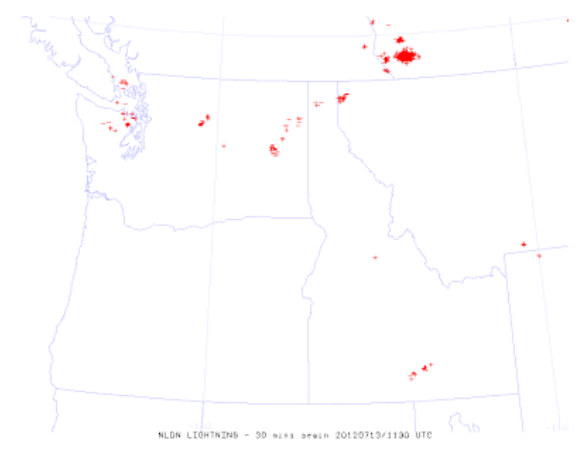


And there are some samples of lightning reports...and folks, there has been huge amounts of lightning for this area.

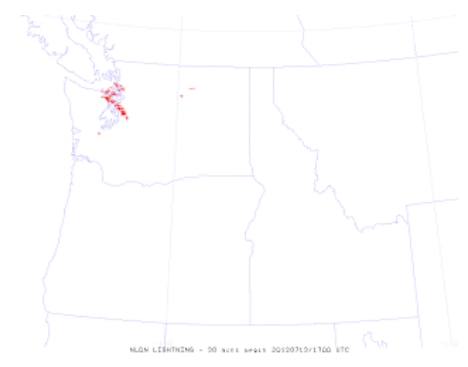
First, for the half-hour ending 11 PM, Thursday....loads in eastern WA and some over the NW Olympic Peninsula



Ending 4:30 AM this AM....eastern Washington and lots over and near the Olympics



And this morning before 10 AM....an active line of thunderstorms over Puget Sound.



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Posted by Cliff Mass at 9:47 AM



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29 comments:

**Kevin Purcell said...**

Wow ... max reflectivity 61dBz on the Camano radar and I can see one cell heading for UW campus across Lake Washington with echo tops at 35k feet. A bit bigger than our usual.

You can hear the thunder from the line of cells on Capitol Hill (first as distant rumbles sounding rather like artillery then real thunder) though most of the action is missing us.

[July 13, 2012 at 10:05 AM](#)

**Beth Niquette said...**

That is the most amazing picture. I adore thunderstorms. However, here in the Mid-Willamette Valley, they usually stay close to the Cascades. We are closer to the Coastal range, so do not see them much.

I do so wish for a thunderstorm. (sigh) Wonderful blog you have here!

[July 13, 2012 at 10:16 AM](#)

**dale said...**

I've lived in Seattle for 5 years now, and this is the first stormy summer morning that's reminded me of my native Massachusetts, where such events are far more commonplace. I've missed them. They are pretty wonderful.

Thanks so much for the post to anticipate and explain what was about to happen, Professor Mass!

[July 13, 2012 at 10:42 AM](#)

**Buddy said...**

Biggest perk of the summer season, thunderstorms. We've really been blessed here in the PNW this year. Wide variety of interesting weather as oppose to the horrid heat and drought consuming in entire country.

[July 13, 2012 at 11:42 AM](#)

**Ferdi said...**

Great thunderstorm passed to the west of Sinclair Island during the night. Spectacular cloud to ground lightening. Apparently lightening set off a small brush fire on Turtleback Mt. on Orcas Island. Good thing June was so wet.

[July 13, 2012 at 12:14 PM](#)

**Justin Sweet said...**

I hate to be a complainer, but where was the NWS on the storms that moved over Seattle today? This morning's forecast didn't even mention a chance of rain--though a chance of thunderstorms was forecast for Saturday. Not the finest hour for the NWS.

[July 13, 2012 at 1:45 PM](#)

**Ben Kessler said...**

Cliff - can you clarify what "onshore flow through depth" means? I understand onshore flow, but the "through depth" part has me puzzled.

Thanks!

[July 13, 2012 at 2:55 PM](#)

**Scott Sullivan said...**

Port Angeles was under some intense lightning bolt strikes at dawn. I shot some amazing photos of some strikes out here. The best I have seen in years. It was reminiscent of a classic east coast storm and the humidity in the air felt like I was somewhere else more tropical. Amazing morning. How can we submit photos to Cliff?

[July 13, 2012 at 3:02 PM](#)

**Upupaepops said...**

so the notion of going over to eastern WA around Thorp to catch possible Aurora Saturday night sound like a potential bust? Thunderstorms vs solar storm

[July 13, 2012 at 4:47 PM](#)

 Hindu said...

I like the Tweet density map. Clever. Just need to weight for population density to balance the map.

[July 13, 2012 at 5:16 PM](#)

 Ferdi said...

7pm and a very active thunderstorm is bearing down on Sinclair Island. So strange to have weather coming from the east. This should be interesting!

[July 13, 2012 at 7:01 PM](#)

 Catherine said...

Cliff, our family is vacationing on San Juan Island. Thunder and lightning almost all day today, which is very strange. But the most unusual phenomenon was the electricity in the air on the beach. We went to picnic at South Beach, which is part of the American Camp National Park. There was fog, with thunder and lightning in the distance across the Strait of Juan de Fuca. When we stepped onto the beach, we all felt a tingling sensation around our scalp, and people's hair started standing on end. Lots of folks on the beach commented on the same phenomenon/the same sensation. What caused this? Is there any danger in being on the beach? Any info you can provide would be appreciated!

[July 13, 2012 at 7:47 PM](#)

**Geoff in Bellevue said...**

Glad to see the report of a LOW down by Olympia. I was watching the animated radar today from Bellevue around 12:00 and saw a line of thunderstorms heading from Issaquah toward the northwest -- and I said "So it looks like we're sitting on the top of a low that's spinning counterclockwise" - and that's exactly what was happening.

[July 13, 2012 at 7:56 PM](#)

 Gary said...

A slow moving band of thunderstorms passed through Olympia between 7:30-8:30PM, with a little bit of big rain splats. It came from the NE and seemed to turn west as it passed over.

[July 13, 2012 at 9:06 PM](#)

 Unknown said...

when your hair stands up that means that the air is charged and your positively charged hair is trying to connect with the electrons in the storm. That means get to cover Lightning is going to strike. The same incident happened to me on the soccer field a few years back. My coach new about this phenomena and told us to get close to the school out of the open and we ran. Right as everyone made it a bolt struck about 400 feet away.

[July 13, 2012 at 9:12 PM](#)

**Robert Sutton said...**

Lots of strikes in the San Juan's tonight. No power.!

[July 13, 2012 at 9:34 PM](#)

 Corie said...

Wheowweeeee....The NOAA station at the Oly airport seems to have been knocked out of commission because no new reports for a couple of hours now. I am eager to find out just how much rain we have had here in West Olympia. Meanwhile, storm #2 is a few miles away. Storm #1 lasted for about an hour, with lots and lots of lightening and thunder, heavy heavy rain and hail, with a beautiful orange sunset coloring everything a deep tangerine color. Amazing!!

[July 13, 2012 at 10:05 PM](#)

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[July 13, 2012 at 10:06 PM](#)

 Gary said...

Update from Olympia: It's almost 10pm and the thunderstorms have continued since they began around 7PM. Now that it's dark, the lightning is very noticeable! And there's almost a continuous background rumbling of thunder; it seems to arrive from many different directions.

We just had our second "reboot worthy" power interruption.

[July 13, 2012 at 10:06 PM](#)

 Kat said...

Catherine,

That is not a good sign. That means there was a static charge building from the ground and you could've been hit by lightning.

"If thunder roars, head indoors"

The NWS has a good info page about lightning, maybe you want to read up on it. There is always a risk when outside during a thunderstorm. Lightning can strike up to 16 miles from the storm.

[July 13, 2012 at 10:22 PM](#)

 slugbiker said...

enjoyed the thunder and lightning show from Queen Anne in Seattle. Here's a short video of a particularly close strike:
<http://youtu.be/hLce9A54LSE>

[July 13, 2012 at 11:01 PM](#)

 Kat said...

Oops! I guess it can strike farther away than I thought.

Here are links for you, Catherine. My facts weren't as accurate as I initially thought.

http://www.srh.noaa.gov/srh/jetstream/lightning/lightning_safety.htm

http://www.srh.noaa.gov/srh/jetstream/lightning/lightning_intro.htm

[July 13, 2012 at 11:11 PM](#)

 epjmcginley said...

Just west of Olympia this evening, heard the sound of lightning before I saw it strike... so close, like aluminum foil being crinkled above my head, and every hair standing on end. I literally found myself clutching the ground in terror. What an amazing feeling.

[July 13, 2012 at 11:55 PM](#)

 KE7DKG said...

It is amazing that I can see the Lightning from these storms in Tacoma all the way down here in Forest Grove, Oregon! There are several bright flashes in the Northern sky every minute!

[July 14, 2012 at 2:20 AM](#)

 Nuthatch said...

Hi! I've read auroras are possible tonight (Saturday.) Do you have any suggestions for where it might be clear in the Puget Sound region?

[July 14, 2012 at 9:11 AM](#)

 habituallydaft said...

Watched last nights lightening activity over the peninsula from a Seattle park with open skies. It was super neat seeing all the contours of the clouds in between here and there getting all lit up with every strike.

It was a neat rare treat.

[July 14, 2012 at 11:07 AM](#)

 Corie said...

I realize weather at home can be vastly different from what NOAA reports. The three day report says that Olympia has "few showers" with a total of .20 inches in six hours. Maybe those were the conditions of the airport, but it rained hard here in West Oly for 4 hours. Does anyone know how I can find the stats for the conditions we experienced?

[July 14, 2012 at 2:10 PM](#)

Dollmaker Barb said...

That thunderstorm lasted for FOUR HOURS down here in Grays Harbor county. One of our dogs barked the entire time; finally the thunder (and the dog) stopped about two a.m. and we could get some sleep!!

[July 14, 2012 at 10:30 PM](#)

 Ansel said...

I was in Marysville Friday morning heading up to Pemberton for a climbing trip when a tremendous lightning bolt was followed within a fraction of a second by a loud bang, and I think it may have struck a tree near the freeway.

Then while in Joffre Lakes Park near Pemberton, there was a show of the Northern lights about midnight, after a hot sunny day of climbing. It was really worth seeing. The following day as we hiked out (Sunday) we had a thunderstorm with light steady rain.

Note to Catherine, Yes, I have experienced these effects, mainly high in the mountains, and it is an early warning: You don't want to be in the open, certainly not in a high spot, and not too near the base of a tree or a cliff. If caught in the open, squat down and make yourself as small as possible. A

canyon is usually safe if you stay away from the creek, which is a path for ground currents. A hard-top car is safe, once you get inside, as it provides a more attractive path for the current. But best to keep away from the windows. If on a sailboat it should be safe if the mast is grounded to the water.

Ansel

July 16, 2012 at 1:41 PM

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